PITWALL F1 COMPLETE TECHNICAL DOCUMENTATION

Generated: 2026-09-13 17:16:07 **Status:** Official Engineering Specification

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2. EXECUTIVE SUMMARY

The Pitwall F1 project is an advanced, telemetry-driven race strategy and visualization dashboard. It ingests FastF1 telemetry data, performs spatial intelligence modeling, evaluates race strategies using Monte Carlo simulations, calculates time loss attribution through differential telemetry math, and presents results in a Streamlit-based UI. The system integrates machine learning for overtake predictions and features an AI-driven 'JARVIS' assistant for querying race parameters. After extensive remediation, it successfully utilizes real telemetry rather than mocked procedural calculations.

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3. PROJECT PURPOSE

To provide Formula 1 racing engineers and strategists with real-time and historical race data analysis, stochastic strategy modeling, and granular spatial telemetry diagnostics in a unified Command Center interface.

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4. SYSTEM ARCHITECTURE

Data Flow Architecture

```
graph TD
    A[User / Streamlit UI] --> B[Pitwall UI (`src/pitwall_ui.py`)]
    B --> C[Telemetry API / Probabilistic Service]
    C --> D[Data Ingestion Engine (FastF1 `src/data_ingestion.py`)]
    C --> E[Strategy & Monte Carlo Engine (`src/monte_carlo_engine.py`)
    C --> F[Telemetry Performance Lab (`src/telemetry_performance_lab.p
    E --> G[Production ML Engine (`src/production_ml_engine.py`)]
    G --> H[Model Loader (`src/model_loader.py`)]
```

Layers

- **Application Entrypoint:** `src/pitwall_ui.py` (Streamlit loop)
- **UI Layer:** `src/pitwall_ui.py`, `src/ui_elements.py`
- **Orchestration Layer:** `src/probabilistic_service.py`, `src/telemetry_api.py`
- **Data Ingestion Layer:** `src/data_ingestion.py` (FastF1 Cache, FIA APIs)
- **Intelligence Layer:**
  - `src/spatial_intelligence_engine.py` (3D Mapping)
  - `src/telemetry_performance_lab.py` (Driver Differentials)

    - `src/unified_tyre_intelligence.py` (Degradation Models)

- **Simulation Layer:** `src/monte_carlo_engine.py`, `src/race_simulator.py`
- **ML Layer:** `src/production_ml_engine.py`, `src/model_loader.py`
- **Persistence/Cache Layer:** Local `fastf1` file cache in `track_cache` and `%APPDATA%`

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5. REPOSITORY STRUCTURE

The project resides primarily within the `src/` directory, supported by a `scripts/` directory for offline jobs like ML training.

```
D:/Pitwall F1 project/
├── src/                    # Core Application logic
├── scripts/                # Offline tools (e.g., train_overtake_model.p
├── models/                 # Serialized ML Models (e.g., model.pkl)
├── track_cache/            # FastF1 offline track metadata
└── docs/                   # Auto-generated documentation
```

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6. COMPLETE FEATURE CATALOGUE

Spatial Intelligence Engine

- **Purpose:** Renders the 3D location of all 20 cars.
- **User capability:** Views real-time track map and car gaps.
- **Inputs:** FastF1 X, Y, Z coordinates.
- **Data source:** FastF1 `lap.get_telemetry()`.
- **Processing:** Extracts telemetry at midpoint, computes dx/dy for heading, and maps coordinates.
- **Models:** None (Direct telemetry).
- **Files:** `spatial_intelligence_engine.py`
- **Status:** PRODUCTION / FUNCTIONAL (Refactored from sine/cosine mocks).

Monte Carlo Strategy Simulator

- **Purpose:** Calculates win probability distributions for pitstop choices.
- **Inputs:** Target compound, pit lap, current grid gaps.
- **Data source:** FastF1 lap time deltas and positions.
- **Processing:** Vectorized numpy simulation sampling from normal/lognormal distributions.
- **Files:** `monte_carlo_engine.py`, `race_simulator.py`
- **Status:** PRODUCTION / FUNCTIONAL

Telemetry Performance Lab

- **Purpose:** Attributes corner time loss.
- **Processing:** Aligns two drivers' arrays by distance, filters by corner bounds, computes deltas for VMin and Peak Brake.
- **Files:** `telemetry_performance_lab.py`
- **Status:** PRODUCTION / FUNCTIONAL

Overtake ML Engine

- **Purpose:** Predicts success probability of an overtake.
- **Processing:** Loads `model.pkl` and infers against gap, DRS, and speed delta.
- **Models:** RandomForestClassifier
- **Files:** `production_ml_engine.py`, `model_loader.py`
- **Status:** PRODUCTION / FUNCTIONAL

JARVIS AI Intelligence

- **Purpose:** AI assistant for pitwall insights.
- **Processing:** Intent routing via Regex/heuristics to internal engine calls.
- **Files:** `jarvis_engineering_insights.py`
- **Status:** EXPERIMENTAL / PARTIALLY FUNCTIONAL

Unified Tyre Intelligence

- **Purpose:** Tyre degradation forecasting.
- **Processing:** Calculates slope of empirical lap times using np.polyfit.
- **Status:** PARTIALLY FUNCTIONAL (Thermal model is heuristic/synthetic)

7. DATA ARCHITECTURE & 8. FASTF1 INGESTION

- **Provider:** FastF1 API
- **Pipeline:** `data_ingestion.py -> fetch_session_data(year, event, session_type)`
- **Cache:** Stored locally in `track_cache` to avoid repeated FIA API hits.
- **Extracted Fields:** Distance, X, Y, Z, Speed, Throttle, Brake, nGear, DRS, Time, LapNumber, Position.

9. TELEMETRY ENGINEERING

- **Time Loss Attribution:** Extracts arrays for Driver A and B, subsets by Corner Distance bounds, calculates `Speed.min()`, `Brake.max()`, and overall time integration over distance. Implementation in `telemetry_performance_lab.py`.

10. STRATEGY ENGINE & 11. MONTE CARLO

- **Simulations:** Default 250 iterations (vectorized).
- **Stochastic Noise:** Uses `np.random.default_rng(seed)` for Thread-Safe Normal (Pace Noise) and Lognormal (Pit Stop Duration Noise).
- **Ranking:** Strategies are ranked by "MAX_WIN_PROBABILITY".

12. MACHINE LEARNING & 13. REINFORCEMENT LEARNING

- **Active ML Model:** `models/overtake_probability/model.pkl` (RandomForestClassifier).
- **Training Pipeline:** `scripts/train_overtake_model.py` generates synthetic overtake data (delta v, DRS state) and trains the scikit-learn model.

- **RL System:** Mentioned in `f1_strategy_env.py` and `rl_strategy_agent.py` but currently **EXPERIMENTAL/ORPHANED** from the main UI orchestration loop.

18. WHAT-IF SCENARIO ENGINE

- **Control:** Sidebar inputs for Compound and Pit Lap.
- **Backend Response:** Injects a custom `StrategyPlan` into the Monte Carlo queue. Overrides the `st.session_state` and invokes `mc_engine.simulate_strategy_vectorized()`. Confirmed functional.

21. DATA PROVENANCE

- **REAL SOURCE DATA:** FastF1 Lap Times, X/Y/Z Telemetry, Position Gaps.
- **DERIVED FROM REAL DATA:** Corner Min Speeds, Brake Points.
- **ML OUTPUT:** Overtake Probability.
- **STATISTICAL CALCULATION:** Win Probability, Expected Finish Position.
- **SYNTHETIC/HEURISTIC:** Tyre Core/Bulk Temperatures.
- **UNAVAILABLE:** Compound Crossover Lap.

27. RUN INSTRUCTIONS

- **Launch Command:** `python -m streamlit run src/pitwall_ui.py`
- **Environment:** Requires `numpy`, `pandas`, `scikit-learn`, `fastf1`, `streamlit`. Python 3.9+ recommended.

29. COMPLETE FILE/FUNCTION REFERENCE

`advanced_monte_carlo.py`

- **Path:** `D:\Pitwall F1 project\src\advanced_monte_carlo.py`
- **Classes:** StrategyCandidate, StrategyMonteCarloResult, AutomatedCandidateGenerator, AdvancedMonteCarloStrategyEngine

`ai_engineer_qa.py`

- **Path:** `D:\Pitwall F1 project\src\ai_engineer_qa.py`
- **Classes:** AIEngineerQAEngine

`ai_engineer_service.py`

- **Path:** `D:\Pitwall F1 project\src\ai_engineer_service.py`
- **Classes:** AIEngineerService

`ai_evaluator.py`

- **Path:** `D:\Pitwall F1 project\src\ai_evaluator.py`
- **Classes:** AIEngineerEvaluator

`ai_tools.py`

- **Path:** `D:\Pitwall F1 project\src\ai_tools.py`
- **Classes:** AIToolRegistry

app.py

- **Path:** `D:\Pitwall F1 project\src\app.py`
- **Functions:** start_live_worker, stop_live_worker, _main_execution_loop, main

authoritative_strategy_engine.py

- **Path:** `D:\Pitwall F1 project\src\authoritative_strategy_engine.py`

bayesian_replay.py

- **Path:** `D:\Pitwall F1 project\src\bayesian_replay.py`
- **Classes:** BayesianRaceReplayEngine

bayesian_state.py

- **Path:** `D:\Pitwall F1 project\src\bayesian_state.py`
- **Classes:** BayesianRaceState

bayesian_tyre_model.py

- **Path:** `D:\Pitwall F1 project\src\bayesian_tyre_model.py`
- **Classes:** AdvancedTyreDegradationModel
- **Functions:** calculate_degradation_slope

bayesian_updating.py

- **Path:** `D:\Pitwall F1 project\src\bayesian_updating.py`
- **Classes:** BayesianParameter, ConjugateBayesianEstimator

benchmark_runner.py

- **Path:** `D:\Pitwall F1 project\src\benchmark_runner.py`
- **Classes:** BenchmarkRunner

cache_service.py

- **Path:** `D:\Pitwall F1 project\src\cache_service.py`
- **Classes:** VersionAwareCache

calibration.py

- **Path:** `D:\Pitwall F1 project\src\calibration.py`
- **Classes:** ProbabilityCalibrationEngine

config.py

- **Path:** `D:\Pitwall F1 project\src\config.py`
- **Functions:** ensure_directories

config_settings.py

- **Path:** `D:\Pitwall F1 project\src\config_settings.py`
- **Classes:** AppConfig

corner_detection.py

- **Path:** `D:\Pitwall F1 project\src\corner_detection.py`
- **Classes:** CornerWindow
- **Functions:** detect_corners_from_telemetry

database_migrations.py

- **Path:** `D:\Pitwall F1 project\src\database_migrations.py`
- **Classes:** DatabaseManager

data_ingestion.py

- **Path:** `D:\Pitwall F1 project\src\data_ingestion.py`
- **Functions:** safe_extract, get_calendar, fetch_session_data, fetch_live_session_data, get_driver_telemetry, get_auto_live_session

data_providers.py

- **Path:** `D:\Pitwall F1 project\src\data_providers.py`
- **Classes:** BaseDataProvider, FastF1Provider, OpenF1Provider
- **Functions:** get_provider

data_sanity.py

- **Path:** `D:\Pitwall F1 project\src\data_sanity.py`
- **Classes:** SanityViolation, DataSanityEngine

decision_architecture.py

- **Path:** `D:\Pitwall F1 project\src\decision_architecture.py`
- **Classes:** DriverContext, RaceContext, StrategyRecommendation, DecisionEngine, JarvisEvidenceEngine, DecisionArchitectureOrchestrator

decision_engine.py

- **Path:** `D:\Pitwall F1 project\src\decision_engine.py`
- **Classes:** DecisionConstraint, DecisionAlternative, StructuredDecision, DecisionEngine

decision_history.py

- **Path:** `D:\Pitwall F1 project\src\decision_history.py`
- **Classes:** HumanOverrideRecord, DecisionAuditTrail

decision_validation.py

- **Path:** `D:\Pitwall F1 project\src\decision_validation.py`

- **Classes:** DecisionValidationEngine

`energy_analysis.py`

- **Path:** `D:\Pitwall F1 project\src\energy_analysis.py`
- **Classes:** SourceClassification, EnergyDataContract, DRSZoneAnalysisResult, DRSAnalysisEngine, ERSBatteryModelEngine

`evidence_engine.py`

- **Path:** `D:\Pitwall F1 project\src\evidence_engine.py`
- **Classes:** EvidenceItem, StructuredEvidenceResponse, EvidenceEngine

`f1_strategy_env.py`

- **Path:** `D:\Pitwall F1 project\src\f1_strategy_env.py`
- **Classes:** F1StrategyEnv

`full_field_intelligence.py`

- **Path:** `D:\Pitwall F1 project\src\full_field_intelligence.py`
- **Classes:** CanonicalDriverAgent, CarState, BattleState, FullFieldDriverState, FullFieldRaceIntelligenceEngine

`full_field_simulator.py`

- **Path:** `D:\Pitwall F1 project\src\full_field_simulator.py`
- **Classes:** DriverSimState, RaceSimulationConfig, FullFieldSimulationResult, FullFieldRaceSimulator

`head_to_head_ai.py`

- **Path:** `D:\Pitwall F1 project\src\head_to_head_ai.py`
- **Functions:** compute_pace_delta, compute_tyre_degradation, compute_telemetry_traits, generate_h2h_report, render_performance_evidence

`historical_ablation.py`

- **Path:** `D:\Pitwall F1 project\src\historical_ablation.py`
- **Classes:** AblationStudyEngine

`historical_backtest.py`

- **Path:** `D:\Pitwall F1 project\src\historical_backtest.py`
- **Functions:** _baseline_strategy_time, _exact_optimal_pit, _rl_episode, backtest_race, main, _make_plots

`historical_leakage_guard.py`

- **Path:** `D:\Pitwall F1 project\src\historical_leakage_guard.py`
- **Classes:** DataLeakageError, HistoricalInformationFilter

historical_metrics.py

- **Path:** `D:\Pitwall F1 project\src\historical_metrics.py`
- **Classes:** HistoricalReplayMetrics

historical_replay_engine.py

- **Path:** `D:\Pitwall F1 project\src\historical_replay_engine.py`
- **Classes:** HistoricalReplayEngine

historical_replay_evaluator.py

- **Path:** `D:\Pitwall F1 project\src\historical_replay_evaluator.py`
- **Classes:** FutureDataGuard, HistoricalReplaySession, DecisionCheckpoint, HistoricalEvaluationReport, HistoricalReplayEngine

historical_retrieval.py

- **Path:** `D:\Pitwall F1 project\src\historical_retrieval.py`
- **Classes:** ContextRelevanceScore, HistoricalRetrievalEngine

jarvis_engineering_insights.py

- **Path:** `D:\Pitwall F1 project\src\jarvis_engineering_insights.py`
- **Classes:** JarvisEngineeringInsightsEngine

job_queue.py

- **Path:** `D:\Pitwall F1 project\src\job_queue.py`
- **Classes:** JobQueueManager

lap_alignment.py

- **Path:** `D:\Pitwall F1 project\src\lap_alignment.py`
- **Functions:** align_laps_by_distance

live_stream_worker.py

- **Path:** `D:\Pitwall F1 project\src\live_stream_worker.py`
- **Functions:** get_oauth_token, on_connect, atomic_write_json, on_message, on_disconnect, start_worker

logger.py

- **Path:** `D:\Pitwall F1 project\src\logger.py`
- **Classes:** JSONFormatter
- **Functions:** sanitize_payload, get_logger

mlops_pipeline.py

- **Path:** `D:\Pitwall F1 project\src\mlops_pipeline.py`
- **Classes:** DatasetVersion, ModelArtifact, DriftMetrics, DatasetRegistry, ModelRegistry,

DriftDetector, ProductionMLOpsEngine

model_loader.py

- **Path:** `D:\Pitwall F1 project\src\model_loader.py`
- **Classes:** ModelLoader
- **Functions:** get_loader

monte_carlo_engine.py

- **Path:** `D:\Pitwall F1 project\src\monte_carlo_engine.py`
- **Classes:** StochasticSimulationOutcome, MonteCarloRaceEngine

observability.py

- **Path:** `D:\Pitwall F1 project\src\observability.py`
- **Classes:** ObservabilityManager

panel_registry.py

- **Path:** `D:\Pitwall F1 project\src\panel_registry.py`
- **Classes:** LayoutPreset, PanelDefinition, PanelRegistry

physics_solver.py

- **Path:** `D:\Pitwall F1 project\src\physics_solver.py`
- **Functions:** render_micro_telemetry_stack, render_pacing_scatter, render_gap_to_leader, render_custom_lap_deficit, render_telemetry_comparison

pitwall_console.py

- **Path:** `D:\Pitwall F1 project\src\pitwall_console.py`
- **Classes:** PitWallConsoleOrchestrator

pitwall_ui.py

- **Path:** `D:\Pitwall F1 project\src\pitwall_ui.py`
- **Functions:** get_tyre_html, format_sector_html, render_session_header, render_engine_component_matrix, render_pitwall_console_header, render_timing_tower, render_track_limits, render_race_control_log, render_driver_time_loss_debrief, render_pipeline_lineage_hud, render_ai_engineer_workstation, render_probabilistic_uncertainty_hud, render_bayesian_updating_hud, render_decision_engine_panel, render_historical_replay_dashboard, render_evaluation_benchmark_dashboard, render_3d_circuit_replay_dashboard, render_unified_realtime_timing_tower, render_universal_session_selector, render_advanced_telemetry_lab, render_drs_ers_energy_dashboard, render_advanced_tyre_degradation_dashboard, render_full_field_simulation_dashboard, render_advanced_monte_carlo_strategy_dashboard, render_historical_backtest_evaluation_dashboard, render_rl_strategy_agent_dashboard, render_ai_engineer_jarvis_console, render_mlops_pipeline_dashboard, render_realtime_live_operations_console, render_driver_performance_intelligence_lab, render_unified_tyre_intelligence_dashboard,

render_authoritative_strategy_command_center, render_full_field_intelligence_dashboard, render_production_ml_model_lab, render_historical_replay_scientific_evaluation_dashboard, render_realtime_live_infrastructure_health_console, render_3d_circuit_spatial_intelligence_dashboard, render_production_data_platform_performance_dashboard, render_production_hardening_security_dashboard, render_system_wide_integration_verification_dashboard, render_master_pitwall_console, render_timing_tower_v2, render_race_control_log_v2, render_weather_panel, render_tyre_panel

probabilistic_model.py

- **Path:** `D:\Pitwall F1 project\src\probabilistic_model.py`
- **Classes:** UncertaintyType, IntervalType, ProbabilisticPrediction
- **Functions:** validate_probability_vector

probabilistic_service.py

- **Path:** `D:\Pitwall F1 project\src\probabilistic_service.py`
- **Classes:** ProbabilisticService

production_data_platform.py

- **Path:** `D:\Pitwall F1 project\src\production_data_platform.py`
- **Classes:** SessionManifest, CacheStats, CircuitBreakerStatus, CircuitBreaker, ProductionCacheManager, SessionPreparationManager

production_hardening_engine.py

- **Path:** `D:\Pitwall F1 project\src\production_hardening_engine.py`
- **Classes:** SecurityAuditReport, HealthCheckResult, BackupManifest, InputSanitizer, RateLimiter, AIBoundaryGuard, FaultToleranceEngine, BackupRestoreManager, SystemHealthService

production_ml_engine.py

- **Path:** `D:\Pitwall F1 project\src\production_ml_engine.py`
- **Classes:** ModelCard, QuantileLapTimePrediction, SHAPExplanation, ProductionMLEngine

race_simulator.py

- **Path:** `D:\Pitwall F1 project\src\race_simulator.py`
- **Classes:** RaceState, DriverState, StrategyOption, SimulationResult, RaceSimulator
- **Functions:** active_inference_strategy_suggestor, render_simulator_ui, render_decision_engine_panel

race_state.py

- **Path:** `D:\Pitwall F1 project\src\race_state.py`
- **Classes:** TyreState, FuelState, WeatherState, TrackState, SafetyCarState, DriverState,

CompleteRaceState

realtime_live_engine.py

- **Path:** `D:\Pitwall F1 project\src\realtime_live_engine.py`
- **Classes:** NormalizedRaceEvent, RaceOperationAlert, EventIdempotencyRegistry, RealtimeAlertEngine, RealtimeRaceOperationsEngine

realtime_live_infrastructure.py

- **Path:** `D:\Pitwall F1 project\src\realtime_live_infrastructure.py`
- **Classes:** NormalizedRaceEvent, LiveSystemHealth, EventIdempotencyRegistry, LiveEventBus, RealtimeLiveInfrastructureEngine

realtime_race_state.py

- **Path:** `D:\Pitwall F1 project\src\realtime_race_state.py`
- **Classes:** RaceControlEvent, DriverRaceState, UnifiedRaceState, BaseDataProvider, ReplayAdapter, OpenF1LiveProvider, RealtimeRaceStateEngine

rl_config.py

- **Path:** `D:\Pitwall F1 project\src\rl_config.py`
- **Functions:** is_rl_enabled

rl_strategy_agent.py

- **Path:** `D:\Pitwall F1 project\src\rl_strategy_agent.py`
- **Classes:** RLObservationState, RLDecisionLog, F1StrategyEnv, RLPolicyAgent, RLStrategyEvaluator

scenario_engine.py

- **Path:** `D:\Pitwall F1 project\src\scenario_engine.py`
- **Classes:** ScenarioPreset, ScenarioEngine

seed_demo_data.py

- **Path:** `D:\Pitwall F1 project\src\seed_demo_data.py`
- **Classes:** DemoDataSeeder

sensitivity_analysis.py

- **Path:** `D:\Pitwall F1 project\src\sensitivity_analysis.py`
- **Classes:** SensitivityAnalysisEngine

spatial_intelligence_engine.py

- **Path:** `D:\Pitwall F1 project\src\spatial_intelligence_engine.py`
- **Classes:** SpatialCarPosition, CockpitPOVCamera, CircuitSpatialMetadata, SpatialIntelligenceEngine

strategy_model.py

- **Path:** `D:\Pitwall F1 project\src\strategy_model.py`
- **Classes:** ObjectiveType, StintPlan, StrategyPlan

strategy_simulator.py

- **Path:** `D:\Pitwall F1 project\src\strategy_simulator.py`
- **Functions:** render_jarvis_deck, render_strategy_timeline

system_evaluation_metrics.py

- **Path:** `D:\Pitwall F1 project\src\system_evaluation_metrics.py`
- **Classes:** MetricWithCI, SystemEvaluationMetrics

system_integration_validator.py

- **Path:** `D:\Pitwall F1 project\src\system_integration_validator.py`
- **Classes:** SubsystemStatus, SystemIntegrationReport, SystemIntegrationValidator

telemetry_analysis_service.py

- **Path:** `D:\Pitwall F1 project\src\telemetry_analysis_service.py`
- **Classes:** TelemetryAnalysisService

telemetry_api.py

- **Path:** `D:\Pitwall F1 project\src\telemetry_api.py`
- **Classes:** TelemetryAPIService

telemetry_cleaning.py

- **Path:** `D:\Pitwall F1 project\src\telemetry_cleaning.py`
- **Functions:** clean_telemetry_df

telemetry_features.py

- **Path:** `D:\Pitwall F1 project\src\telemetry_features.py`
- **Classes:** LapFeatures, CornerFeatures, FeatureDataset, FeatureEngineeringService

telemetry_ingestion.py

- **Path:** `D:\Pitwall F1 project\src\telemetry_ingestion.py`
- **Classes:** IngestedDataset, TelemetryIngestionService

telemetry_lab.py

- **Path:** `D:\Pitwall F1 project\src\telemetry_lab.py`
- **Classes:** CanonicalTelemetryPoint, CornerAnalysisResult, TelemetryQualityEngine, TrackDistanceAligner, TimeLossAttributor

telemetry_model.py

- **Path:** `D:\Pitwall F1 project\src\telemetry_model.py`
- **Classes:** TelemetryUnits, CanonicalTelemetryPoint, CanonicalLapTelemetry

`telemetry_performance_lab.py`

- **Path:** `D:\Pitwall F1 project\src\telemetry_performance_lab.py`
- **Classes:** CanonicalTelemetrySample, CornerDefinition, CornerTimeLossAttribution, PerformanceOpportunity, DriverPerformanceProfile, DriverPerformanceIntelligenceEngine

`telemetry_pipeline.py`

- **Path:** `D:\Pitwall F1 project\src\telemetry_pipeline.py`
- **Classes:** PipelineRunResult, TelemetryPipeline

`telemetry_schema.py`

- **Path:** `D:\Pitwall F1 project\src\telemetry_schema.py`
- **Classes:** FieldDefinition
- **Functions:** validate_schema_compliance

`telemetry_storage.py`

- **Path:** `D:\Pitwall F1 project\src\telemetry_storage.py`
- **Classes:** TelemetryStorageEngine

`telemetry_sync.py`

- **Path:** `D:\Pitwall F1 project\src\telemetry_sync.py`
- **Functions:** synchronize_telemetry_channels

`telemetry_validation.py`

- **Path:** `D:\Pitwall F1 project\src\telemetry_validation.py`
- **Classes:** TelemetryValidationResult
- **Functions:** validate_telemetry_df

`test_model_loader.py`

- **Path:** `D:\Pitwall F1 project\src\test_model_loader.py`
- **Functions:** test_all_models

`three_d_replay_engine.py`

- **Path:** `D:\Pitwall F1 project\src\three_d_replay_engine.py`
- **Classes:** CameraMode, Car3DState, ThreeDTrackCoordinateSystem, ReplayCacheEngine, ThreeDSceneGenerator

`time_loss_attribution.py`

- **Path:** `D:\Pitwall F1 project\src\time_loss_attribution.py`
- **Classes:** CornerPerformanceAnalysis

- **Functions:** analyze_corner_performance, _build_empty_corner_analysis

tracinginsights.py

- **Path:** `D:\Pitwall F1 project\src\tracinginsights.py`
- **Functions:** get_driver_color, calculate_fuel_corrected_pace, apply_tyre_smoothing, generate_fuel_pace_chart, generate_25_zone_map, generate_race_trace, generate_pace_heatmap, detect_corners_from_speed, generate_cornering_kinematics, validate_z_axis, fetch_openf1_intervals, fetch_openf1_radio, map_radio_to_laps, generate_radio_timeline, fetch_fia_documents, render_pdf_viewer

track_geometry.py

- **Path:** `D:\Pitwall F1 project\src\track_geometry.py`
- **Classes:** TrackGeometryCache
- **Functions:** load_track_geometry, ensure_track_geometry, render_spatial_map, render_animated_track_trace, render_2d_track_with_positions

train_lap_time_model.py

- **Path:** `D:\Pitwall F1 project\src\train_lap_time_model.py`
- **Functions:** get_lap_data, main

train_overtake_model.py

- **Path:** `D:\Pitwall F1 project\src\train_overtake_model.py`
- **Functions:** train_overtake_model

train_rl_agent.py

- **Path:** `D:\Pitwall F1 project\src\train_rl_agent.py`
- **Functions:** main

trend_analysis.py

- **Path:** `D:\Pitwall F1 project\src\trend_analysis.py`
- **Classes:** TimeSeriesTrendEngine

tyre_engine.py

- **Path:** `D:\Pitwall F1 project\src\tyre_engine.py`
- **Classes:** CanonicalTyreObservation, StintMetadata, TyrePredictionResult, EmpiricalTyreModel, PhysicsInformedTyreModel, TyreModelEvaluator

ui_elements.py

- **Path:** `D:\Pitwall F1 project\src\ui_elements.py`
- **Functions:** inject_custom_css, render_standby_screen, render_live_track_map, inject_hotkeys

unified_tyre_intelligence.py

- **Path:** `D:\Pitwall F1 project\src\unified_tyre_intelligence.py`
- **Classes:** CanonicalTyreState, CompoundCrossoverForecast, PitWindowForecast, UnifiedTyreIntelligenceEngine

`universal_session_engine.py`

- **Path:** `D:\Pitwall F1 project\src\universal_session_engine.py`
- **Classes:** EventSessionInfo, UniversalSessionEngine, SessionPreparationManager

`websocket_manager.py`

- **Path:** `D:\Pitwall F1 project\src\websocket_manager.py`
- **Classes:** WebSocketConnectionManager

`__init__.py`

- **Path:** `D:\Pitwall F1 project\src\__init__.py`